BLOOD MANAGEMENT SYSTEM

Detailed Feature Specification

Saving Lives Through Technology

A Comprehensive Multi-Role Platform for India's Blood Supply Chain

Product	Blood Management System (BMS)
Scope	Hospital · Blood Bank · Rider · Public Portal
Tech Stack	React.js · Java Spring Boot · MySQL · Flutter
Cloud	AWS / Azure / DigitalOcean
Status	Feature Design Phase

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⚕ PROBLEM STATEMENT

Why India Needs a Blood Management System

India's healthcare system faces a critical gap in blood supply chain management. In emergencies, every minute counts — yet hospitals routinely spend precious time making phone calls to locate the right blood type, in the right quantity, from the right bank.

#	Problem	Impact on Patient Care
1	**No Centralized System**	Hospitals cannot check real-time blood availability across banks — staff rely on phone calls and personal contacts.
2	**Manual Search Process**	Doctors manually call multiple blood banks in emergencies, wasting critical minutes.
3	**Unorganized Transport**	No structured rider/logistics system — blood delivery is ad-hoc, risking cold-chain breaks.
4	**Delayed Response**	Delay in locating and delivering blood leads directly to loss of patient life.
5	**No Expiry Tracking**	Blood units expire unused due to poor inventory visibility across the supply chain.
6	**Rare Blood Groups**	Rare blood types (e.g., AB−, O−) are especially hard to locate without a centralized registry.

The Blood Management System (BMS) directly addresses each of these pain points with a real-time, centralized, multi-role digital platform.

👥 USER ROLES & ACCESS CONTROL

Multi-Login System with Role-Based Permissions

The BMS operates on a strict Role-Based Access Control (RBAC) model. Each user type accesses only the features relevant to their responsibilities, ensuring security, clarity, and accountability at every level.

User Role	Dashboard Access & Permissions
Admin (Super User)	Full system control – users, banks, hospitals, riders, reports, monitoring
Blood Bank Manager	Manage inventory, view requests, accept/reject, assign riders, blood camp management
Hospital / Doctor	Search availability, raise requests, track delivery, view usage analytics
Rider / Delivery	View assigned tasks, navigate pickup/drop, update status, OTP confirmation
Donor (Future)	Register profile, track donation history, receive camp notifications

All roles authenticate via JWT-secured APIs. OTP-based login is supported for hospital and rider accounts. Admin retains override capabilities across all modules.

HOSPITAL / DOCTOR DASHBOARD

Request Blood · Track Delivery · Manage Usage

The Hospital Dashboard is designed for speed and clarity. Doctors and hospital admins can search for blood, submit requests, track deliveries in real time, and review historical usage — all from a single, intuitive interface.

1. Login & Authentication

•	**Login Methods**	Secure login via OTP (mobile-based) or Password credential
•	**Session Security**	JWT token-based session management with auto-expiry
•	**Role Detection**	System auto-routes to Hospital Dashboard on successful authentication
•	**Password Recovery**	OTP-based reset sent to registered mobile or email

2. Blood Availability Search

•	**Blood Group Filter**	Select from all 8 groups: A+, A−, B+, B−, AB+, AB−, O+, O−
•	**Quantity Input**	Enter required units (e.g., 2 units of B+ Whole Blood)
•	**Component Type**	Filter by: Whole Blood, PCV (Packed Cell Volume), FFP, SDP (Platelet)
•	**Nearest Banks**	System displays nearest blood banks with sufficient stock, ranked by distance
•	**Live Stock Status**	Real-time availability counts pulled from blood bank inventory
•	**Map View**	Blood banks shown on embedded Google Maps for visual selection

3. Raise Blood Request

•	**Patient Details**	Patient name, age, ward/bed number, attending physician
•	**Blood Details**	Blood group, component type, quantity, required-by date/time
•	**Urgency Level**	Normal / Urgent / Critical — affects prioritization and alert type
•	**Target Bank**	Select from search results or auto-assign to best available bank
•	**Notes Field**	Optional clinical notes or special handling instructions for blood bank
•	**Submission**	One-click submit with instant notification sent to selected blood bank

4. Request Tracking

•	**Status: Pending**	Request submitted, awaiting blood bank review and acceptance

•	**Status: Accepted**	Blood bank confirmed stock and approved the request
•	**Status: Rejected**	Blood bank unable to fulfill — reason displayed, alternative suggested
•	**Status: In Transit**	Rider picked up and is en route — live GPS tracking active
•	**Status: Delivered**	Hospital confirmed receipt — request closed and archived
•	**Live ETA**	Estimated arrival time shown with real-time rider map tracking

5. Notifications & Alerts

•	**Accept/Reject Alert**	Instant push notification when blood bank accepts or rejects request
•	**Dispatch Alert**	Notification when rider is assigned and blood has been picked up
•	**Delivery Alert**	Push and in-app notification on confirmed delivery
•	**Expiry Warning**	Alert if a matched unit is approaching expiry and action is needed
•	**Emergency Broadcast**	Critical requests trigger high-priority alerts to blood bank staff
•	**Channels**	In-app banner, email, and FCM push notification (configurable)

6. History & Analytics Dashboard

•	**Usage Timeline**	Monthly and yearly blood usage charts per hospital
•	**Request Log**	Full searchable history of all requests with status, date, and fulfillment time
•	**Blood Group Stats**	Breakdown of most requested blood groups over time
•	**Fulfillment Rate**	Percentage of requests fulfilled, rejected, and average delivery time
•	**Export Reports**	Download usage reports as PDF or Excel for procurement planning
•	**Trend Analysis**	Identify seasonal demand spikes for proactive blood camp planning

7. Emergency Mode

•	**Activation**	Doctor toggles 'Emergency Mode' when submitting a critical request
•	**Highlighting**	Request flagged with RED URGENT banner visible to all blood bank staff
•	**Priority Queue**	Emergency requests jump to top of blood bank request queue
•	**Rare Blood Alert**	System auto-broadcasts to all banks within 50 km for rare groups (AB−, O−, etc.)

•	**SMS Fallback**	Critical alerts sent via SMS in addition to push — ensures delivery even with poor internet
•	**Audit Trail**	Emergency activations logged with timestamp and authorizing doctor's credentials

BLOOD BANK / ADMIN DASHBOARD

Inventory · Requests · Donors · Analytics

The Blood Bank Dashboard is the operational core of the BMS. Bank managers control inventory, manage incoming hospital requests, coordinate rider assignments, run blood camps, and generate comprehensive reports.

1. Blood Inventory Management

•	**Add Blood Units**	Log each donated unit with full metadata: blood group, component type, volume (ml)
•	**Collection Date**	Date and time of collection recorded for freshness tracking
•	**Expiry Date**	System auto-calculates expiry based on component type; alerts at 7-day and 1-day thresholds
•	**Blood Components**	Categorize as: Whole Blood, PCV (Packed Cell Volume), FFP (Fresh Frozen Plasma), SDP (Single Donor Platelet)
•	**Unit ID / Barcode**	Each unit assigned unique ID for full traceability from donation to delivery
•	**Batch Operations**	Bulk upload units from camp collection sheets via CSV import

2. Real-Time Stock Dashboard

•	**Live Counts**	Current available units per blood group and component type, updated in real time
•	**Auto-Decrement**	Inventory automatically reduced when a request is accepted and rider assigned
•	**Low-Stock Alerts**	Configurable threshold alerts (e.g., notify when O+ drops below 5 units)
•	**Expiry Warnings**	Units approaching expiry highlighted in amber/red; action prompts shown
•	**Visual Dashboard**	Color-coded heatmap view of stock levels across all blood groups
•	**Reserved vs. Available**	Distinction between units available and those soft-reserved for accepted requests

3. Hospital Request Management

•	**Incoming Queue**	All hospital requests shown in real-time with blood group, quantity, urgency, and hospital name
•	**Accept Request**	One-click accept — inventory soft-reserved, hospital notified instantly
•	**Reject Request**	Reject with mandatory reason selection; hospital receives immediate notification
•	**Assign Rider**	Post-acceptance, assign available rider from registered rider pool

•	**Priority Sorting**	Critical and Emergency requests appear at top of queue with visual flags
•	**Request History**	Full log of all handled requests with outcomes, timestamps, and rider details

4. Donor Management

•	**Donor Profiles**	Name, contact, blood group, date of birth, health questionnaire responses
•	**Donation History**	Full log of each donation: date, unit volume, camp or walk-in, health status at time
•	**Eligibility Check**	System flags donors not yet eligible to re-donate (based on 56-day / 90-day rules)
•	**Recall System**	Notify eligible donors for upcoming camps or urgent rare-group needs via SMS/app
•	**Donor ID Card**	Printable/digital donor ID with QR code linking to donation history
•	**Search & Filter**	Search donors by blood group, location, last donation date, or eligibility status

5. Blood Camp Management

•	**Create Campaign**	Set up camp with name, location (GPS + address), date, time, and target collection goal
•	**Partner Institutions**	Link camp to organizing hospitals, colleges, or corporates
•	**Donor Registration**	Pre-registration via public portal; walk-in registration at camp kiosk
•	**Real-Time Collection**	Track units collected during camp with live running total
•	**Post-Camp Upload**	Batch upload all collected units to inventory with camp-linked metadata
•	**Camp Analytics**	Per-camp report: donors attended, units collected, blood group breakdown, target vs. actual

6. Reports & Analytics

•	**Collection Reports**	Monthly / quarterly units collected by blood group, component, and source (camp vs. walk-in)
•	**Supply Reports**	Units dispatched, delivered, rejected, and expired per period
•	**Hospital-Wise Data**	Which hospitals requested most, fulfilment rates, and average response time
•	**Rider Performance**	Delivery success rate, average delivery time, and on-time percentage per rider
•	**Wastage Reports**	Units expired before use — helps optimize intake vs. demand forecasting

•	**Export Options**	PDF, Excel, and CSV exports for regulatory compliance and internal review

7. Logistics Dashboard

•	**Active Deliveries**	Real-time view of all ongoing deliveries with rider location on map
•	**Rider Status**	Available, On Pickup, In Transit, Completed — updated automatically from rider app
•	**ETA Monitoring**	Estimated time for each active delivery; alerts if delivery exceeds expected window
•	**Delivery History**	Full log of all past deliveries with timestamps, rider, hospital, and confirmation method
•	**Incident Logging**	Log delivery issues: cold chain breach, damage, delay, or refusal — linked to request ID
•	**Route Playback**	Replay GPS route of any completed delivery for audit and training purposes

🛵 RIDER / DELIVERY APP

Pickup · Navigation · Delivery Confirmation

The Rider App is a lightweight mobile application (Flutter/React Native) designed for real-world field use. It gives delivery riders clear task management, navigation, and status updating tools to ensure fast, reliable, and documented blood deliveries.

1. Login & Profile (Mobile)

•	**Mobile OTP Login**	Rider logs in via registered mobile number + OTP — no password required
•	**Profile Details**	Name, vehicle type, license plate, assigned zone, and blood bank affiliation
•	**Availability Toggle**	Rider marks themselves as Available or Off-Duty — affects assignment eligibility
•	**Device Registration**	Device linked to account for secure push notifications and location tracking

2. Task Assignment & Notification

•	**Push Notification**	Instant alert when blood bank assigns a pickup task — includes blood type, hospital name, and urgency
•	**Task Details Screen**	Full task info: blood unit ID, pickup address, destination hospital, contact person
•	**Accept / Decline**	Rider can accept or decline; declined tasks immediately re-assigned to next available rider
•	**Task Queue**	Active and upcoming tasks shown in a prioritized list with urgency indicators

3. Navigation & Maps

•	**Google Maps Integration**	Turn-by-turn navigation from rider's current location to blood bank (pickup) and then to hospital (drop)
•	**Optimized Routing**	System suggests fastest route accounting for real-time traffic
•	**Cold Storage Points**	Map shows pre-registered cold storage exchange points for multi-leg deliveries
•	**Offline Maps**	Partial offline support for areas with low connectivity

4. Pickup Workflow

•	**Arrive at Bank**	Rider taps 'I Have Arrived' at blood bank — timestamp logged
•	**Unit Verification**	Scan unit barcode or enter unit ID to confirm correct blood unit collected

•	**Document Collection**	Confirm receipt of transport documents (cross-match report, chain of custody form)
•	**Cold Chain Check**	Rider confirms cold storage packaging is intact before marking 'Picked Up'
•	**Pickup Confirmation**	Blood bank receives confirmation; inventory status moves to 'In Transit'

5. Delivery Workflow

•	**Arrive at Hospital**	Rider taps 'Arrived at Hospital' — hospital staff notified to receive
•	**OTP Confirmation**	Hospital staff receives OTP; rider enters OTP in app to confirm handover
•	**Signature Option**	Alternative to OTP: digital signature capture on rider's mobile device
•	**Condition Report**	Rider marks condition of unit on delivery: Good / Temperature Alert / Damaged
•	**Delivery Complete**	System marks request as Delivered; inventory decremented; report auto-generated

6. Live Location Tracking

•	**GPS Broadcasting**	Rider app streams GPS coordinates every 10 seconds while task is active
•	**Visible To**	Blood bank logistics dashboard and hospital tracking screen both show live rider location
•	**Privacy Control**	Location tracking only active during an assigned task — off when idle or off-duty
•	**ETA Updates**	ETA recalculated dynamically based on rider's current position and traffic

7. Status Update System

•	**Assigned**	Task created and assigned to rider — awaiting acceptance
•	**Accepted**	Rider confirmed task — heading to blood bank for pickup
•	**Picked Up**	Blood unit collected and verified at blood bank
•	**In Transit**	Rider en route to hospital — live GPS active
•	**Delivered**	Hospital confirmed receipt via OTP or signature
•	**Issue Reported**	Rider flagged a delivery issue (delay, cold chain, refusal) for blood bank review

⚙ ADMIN (SUPER USER) PANEL

System-Wide Control & Monitoring

The Admin panel gives a single super-user complete visibility and control over every entity in the system — blood banks, hospitals, riders, donors, and platform health. This role is distinct from the Blood Bank Manager role.

1. Entity Management

•	**Add Blood Banks**	Register new blood banks with name, location, license number, and contact details
•	**Add Hospitals**	Onboard hospitals with admin user creation, department details, and contact info
•	**Add Riders**	Register delivery riders with vehicle details and zone assignment
•	**User Management**	Create, activate, suspend, or delete any user account across all roles
•	**Role Assignment**	Assign and modify user roles; promote blood bank staff to manager level

2. System Monitoring

•	**Platform Health**	Real-time uptime, API response times, active sessions, and error rate dashboard
•	**Active Requests**	Live view of all open blood requests across every hospital and blood bank
•	**Rider Activity**	Map view of all active riders with their current delivery status
•	**Inventory Overview**	Aggregated national view of total blood units by type across all registered banks
•	**Audit Logs**	Full immutable log of every action: logins, inventory changes, request decisions, and role changes

3. Reports & Compliance

•	**National Reports**	Aggregate collection, supply, wastage, and fulfillment data across all banks and hospitals
•	**Bank Performance**	Compare banks by fulfillment rate, response time, and inventory turnover
•	**Compliance Export**	Generate CDSCO/Drug Controller-ready reports for regulatory submissions
•	**Alert Configuration**	Set system-wide thresholds for low-stock alerts, expiry warnings, and SLA breach notifications

🔄 END-TO-END SYSTEM FLOW

From Donation to Delivery — 10 Steps

The following workflow describes the complete lifecycle of a blood unit within the BMS — from initial donation at a camp, through inventory management, hospital request, rider dispatch, and final delivery confirmation.

STEP01	**Blood Collection**	Donor visits blood camp → Sample tested → Approved units stored in blood bank with full metadata (type, date, volume).
STEP02	**Inventory Update**	Blood bank staff enters units into the BMS system, categorized by blood group, component (PCV/FFP/SDP), collection date, and expiry.
STEP03	**Hospital Request**	Doctor logs in → Selects blood group + quantity → Adds patient details and urgency level → Submits request.
STEP04	**Smart Matching**	System geographically filters blood banks with sufficient available stock, ranks by proximity and availability score.
STEP05	**Blood Bank Action**	Blood bank receives notification → Reviews request details → Accepts or rejects with reason → Inventory soft-reserved on acceptance.
STEP06	**Rider Assignment**	Blood bank assigns an available rider to the approved request. Rider receives push notification with pickup details.
STEP07	**Pickup & Documentation**	Rider arrives at blood bank → Verifies unit → Collects cold-chain packaging + transport documents → Marks 'Picked Up' in app.
STEP08	**Cold-Chain Delivery**	Blood transported in temperature-controlled container. Real-time GPS tracking visible to hospital and blood bank.
STEP09	**Hospital Confirmation**	Rider arrives → Hospital staff confirms receipt via OTP or digital signature → Status updated to 'Delivered'.
STEP10	**System Update**	Inventory decremented, request closed, delivery report generated. Data feeds into monthly analytics and audit trail.

💻 TECHNOLOGY STACK & SECURITY

Architecture · APIs · Security Features

The BMS is built on a modern, scalable, and secure technology stack. Each component is selected for reliability in low-bandwidth Indian healthcare environments, with cloud hosting ensuring high availability.

Layer	Technology	Purpose / Details
Frontend	**React.js + Tailwind CSS**	Responsive dashboards for Hospital & Blood Bank
Backend	**Java Spring Boot**	RESTful APIs, business logic, service orchestration
Database	**MySQL**	Relational data: inventory, requests, users, audit logs
Mobile App	**Flutter / React Native**	Rider delivery app (Android & iOS)
Maps	**Google Maps API**	Blood bank proximity search, rider navigation
Notifications	**Firebase Cloud Messaging**	Push alerts for hospitals and riders
Real-Time	**Socket.io**	Live tracking, instant request status updates
Cloud	**AWS / Azure / DigitalOcean**	Hosting, storage, scaling, disaster recovery
Auth	**JWT + OTP**	Secure token-based login with mobile OTP fallback
Security	**RBAC + AES Encryption**	Role-based access control, data encryption, audit logs

Security Features

1	Role-Based Access Control (RBAC) — users can only access features assigned to their role
2	JWT (JSON Web Token) authentication for all API endpoints with short-lived token expiry
3	OTP-based login for hospital and rider accounts — eliminates password theft risk
4	AES-256 data encryption at rest for all patient and donor personal information
5	TLS/HTTPS encryption in transit for all API communication
6	Comprehensive audit logs — every action logged with timestamp, user ID, and IP address
7	SQL injection and XSS prevention via input validation and parameterized queries
8	Rate limiting on all API endpoints to prevent abuse and DDoS attacks
9	Cold chain data signed with checksum to detect tampering during transport
10	GDPR/DPDP-aligned data handling with consent management for donor profiles

Built to Save Lives — One Drop at a Time

For implementation details, API documentation, or demo access — contact the project team.